



Maximum Matchings

Lecture 7 | Algorithms Design & Analysis

Girish Varma
IIIT Hyderabad

Recall: Gale-Shapley Algorithm



Repeat until all men are matched

1. Each woman proposes to most preferred man
2. Each man tentatively chooses the most preferred woman from whom he got proposals and rejects the others.
3. If a man who was tentatively matched previously, gets a better offer according to his preference list, he switches.
4. Woman who gets reject, removes that man from their preference list.

Optimal for Woman



Theorem: Algorithm results in every woman getting her best choice among all possible stable matchings.

Theorem: Algorithm results in every man getting his worst choice among all possible stable matchings.

Proofs: Page 10 in Klienberg-Tardos:

<https://theswissbay.ch/pdf/Gentoomen%20Library/Algorithms/Algorithm%20Design%20-%20John%20Kleinberg%20-%20%C3%89va%20Tardos.pdf#page=13.40>

Read More: Popular Matchings



Watch: <https://www.youtube.com/watch?v=nbaRTJtYwuk>

Hall's Theorem



Theorem 2. *For a bipartite graph G on the parts X and Y , the following conditions are equivalent.*

- (a) *There is a perfect matching of X into Y .*
- (b) *For each $T \subseteq X$, the inequality $|T| \leq |N_G(T)|$ holds.*

Proof:

a.) $\Rightarrow$ b.) : Easy

b.) $\Rightarrow$ a.) : 2 Cases

Case 1: b is satisfied with strict inequality.

Easy

Case 2: There is some T for which b is satisfied only with equality

See: <https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>

Idea: Augmenting Paths



Definition 3. If G is a graph and M is a matching in G , a vertex is called *matched* if it belongs to one of the edges in M , and *free* otherwise.

An *alternating component with respect to M* (also called an *M -alternating component*) is an edge set that forms a connected subgraph of G of maximum degree 2 (i.e., a path or cycle), in which every degree-2 vertex belongs to exactly one edge of M . An *augmenting path with respect to M* is an M -alternating component which is a path both of whose endpoints are free vertices.

Idea: Augmenting Paths



In the following lemma, and throughout these notes, we use the notation $A \oplus B$ to denote the *symmetric difference* of two sets A and B , i.e. the set of all elements that belong to one of the sets but not the other.

Lemma 1. *If M is a matching and P is an augmenting path with respect to M , then $M \oplus P$ is a matching containing one more edge than M .*

Lemma 2. *A matching M in a graph G is a maximum cardinality matching if and only if it has no augmenting path.*

Algorithm 1 Naïve iterative scheme for computing a maximum matching

- 1: Initialize $M = \emptyset$.
 - 2: **repeat**
 - 3: Find an augmenting path P with respect to M .
 - 4: $M \leftarrow M \oplus P$
 - 5: **until** there is no augmenting with respect to M .
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How to find Augmenting Paths?



Definition 4. If $G = (L, R, E)$ is a bipartite graph and M is a matching, the graph G_M is the directed graph formed from G by orienting each edge from L to R if it does not belong to M , and from R to L otherwise.

Lemma 3. Suppose M is a matching in a bipartite graph G , and let F denote the set of free vertices. M -augmenting paths are in one-to-one correspondence with directed paths from $L \cap F$ to $R \cap F$ in G_M .

Read



For Stable Matchings
Klienber Tardos, Chapter 1

For Hall's Theorem Proof:

<https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>

For Bipartite Matchings, Section 1 in

<https://www.cs.cornell.edu/courses/cs6820/2023fa/handouts/matchings.pdf>

Solve



Exercises 1, 2 in

<https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>